

Specifications (To design OP Amp)

1) DC gain = 1000 (60 dB) $20 \log_{10} 10^3 \Rightarrow 60 \log_{10} 10^3$

2) GBW = 40 MHz.

3) PM $\geq 45^\circ$

4) VDD = 1.8V

5) Slew Rate = $\frac{20V}{\mu\text{sec}}$

$L_{\min} = 90\text{nm}$
process = 90nm.

6) $ICMR(+)$ max = 1.6V
 $ICMR(-)$ min = 0.8V

$L \geq 2(L_{\min})$

$L \geq 180\text{nm}$

because of λ
channel length
modulation.

7) $C_L = 2\text{PF}$

8) Total power dissipation $\leq 300\text{mW}$.

Let $L = 500\text{nm}$

to avoid
channel length
modulation.

Analysis

for phase margin $\geq 45^\circ$

from Bode plot (at $\omega = \text{GBW}$)

$$-180^\circ + \text{pm} \Rightarrow \tan^{-1}\left(\frac{1}{10}\right) - \underbrace{\tan^{-1}(A_{pc})}_{\rightarrow 90^\circ} - \tan^{-1}\left(\frac{\text{GBW}}{P_2}\right)$$

we assumed

$$Z_1 \geq 10(\text{GBW})$$

very large value for
DC gain

for 45° PM value

Tan can accommodate
very high value in 90°

we get

$$-180^\circ + 45^\circ \Rightarrow \tan^{-1}\left(\frac{1}{10}\right) - 90^\circ - \tan^{-1}\left(\frac{\text{GBW}}{P_2}\right)$$

$$\text{pm} \geq 45^\circ \text{ then } \underline{P_2 \geq 1.22 \text{GBW}} \rightarrow \textcircled{1}$$

$$Z_1 \geq 9(\text{GBW})$$

$$\frac{g_{m2}}{C_c} \geq 9 \left(\frac{g_{m1}}{C_c} \right)$$

$$\underline{g_{m2} \geq 9 g_{m1}} \rightarrow \textcircled{2}$$

from (1)

$$P_d \geq 1.22 \text{ GBW}$$

$$\frac{g_{m2}}{C_L} \geq 1.22 \left(\frac{g_{m1}}{C_C} \right)$$

from (2) put g_{m2} as 10 gm1

$$\frac{10 g_{m1}}{C_L} \geq \left(\frac{1.22 g_{m1}}{C_C} \right)$$

$$\text{WET } C_L = 2 \text{ PF}$$

$$\frac{10}{2 \times 10^{-12} \text{ F}} \geq \frac{1.22}{C_C}$$

$$C_C \geq \frac{2.4 \times 10^{-12} \text{ F}}{10}$$

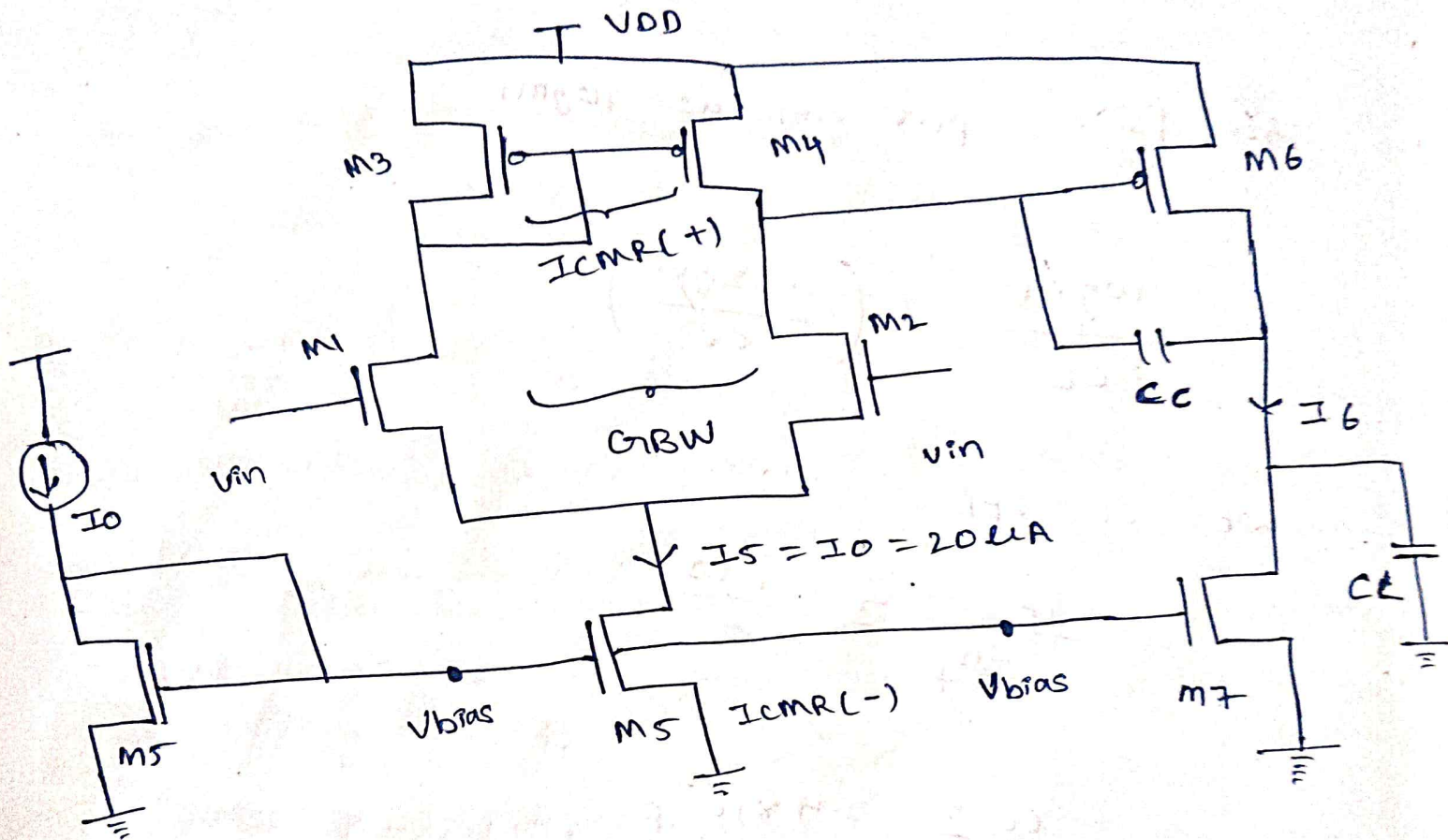
$$C_C \geq 0.24 \text{ PF}$$

(or)

$$C_C \geq 240 \text{ aF}$$

Let C_C be 1 PF

Design



* for m_1 & m_2

$$g_{m1} = GBW (C_c) 2\pi$$

gm, $\Rightarrow 40 \times 10^6 \left(\frac{10^{-12}}{10^{-12}} \right) 6.28$

gm. $\Rightarrow$ 164 u

We know that

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$I_D = \frac{\mu_n C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{th})^2}{2} \quad \text{for saturation}$$

$$\frac{\partial I_D}{\partial V_{GS}} = \frac{\mu_n C_{ox} \left(\frac{W}{L}\right) 2(V_{GS} - V_{th})}{2}$$

$$\frac{\partial I_D}{\partial V_{GS}} = g_m \Rightarrow \mu_n C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{th})$$

$$\frac{g_m^2}{2} = \frac{\left(\mu_n C_{ox} \left(\frac{W}{L}\right)\right)^2 (V_{GS} - V_{th})^2}{2}$$

$$\frac{g_m^2}{2} = \mu_n C_{ox} \left(\frac{W}{L}\right) (I_D)$$

$$\ast \therefore \left(\frac{W}{L}\right) = \frac{g_m^2}{2 I_D (\mu_n C_{ox})}$$

from the circuit, we can understand that

$$I_S = I_D + I_D = 2 I_D$$

$$\left(\frac{W}{L}\right)_{1,2} \Rightarrow \frac{g_{m2}}{2I_D(\text{linloc})}$$

$$\boxed{\left(\frac{W}{L}\right)_{1,2} \approx 6}$$

* for M_3, M_4

M_3 will always be in saturation.

[because gate and drain shorted]

$$V_d \geq V_g - V_{th}$$

$$V_g \leq V_{d1} + V_{th}$$

$$V_{in} \leq V_{d1} + V_{th}$$

$$V_{inmax} = V_{d1} + V_{th}$$

$$\text{use } \mu_{p102} = 60 \mu A/V^2$$

$$k_{n102} = 300 \mu A/V^2$$

$$\left(\frac{W}{L}\right)_{3,4} \Rightarrow \frac{g_{m2}}{2I_D(\mu_{p102})}$$

$$\text{for } L = 500 \text{ nm}$$

$$\left(\frac{W}{L}\right)_{3,4} \approx 14$$

$$270$$

$$\mu_{p102} (V_{DD} - I_{CM2} - V_{3max} + V_{th3})$$

* for m_5

$V_{ds} > V_g - V_t$ (for m_5 to be in saturation)

$$\left(\frac{\omega}{L}\right)_5 \Rightarrow \frac{2I_{D5}}{C_{in(ox)}(V_{Dsat})^2} \quad \text{for } m_5 \text{ to be in saturation.}$$

$$\left(\frac{\omega}{L}\right)_5 \approx 12.$$

* for m_6

for 45° PM

from design.

$$g_{m6} \geq 9g_{m1}$$

$$V_{GS3} = V_{GS4} = V_{GS6}$$

$$g_{m6} \geq 9(160\mu)$$

$$g_{m6} \geq 1.44\text{mA}$$

$$\left(\frac{\omega}{L}\right)_6 = \frac{I_6}{I_4} \Rightarrow \frac{g_{m6}}{g_{m4}}$$

$$* g_{m2} = \frac{2I_D \mu_p C_{ox} \left(\frac{\omega}{L}\right)}{1}$$

$$g_{m4} \Rightarrow \sqrt{98 \times 14 \times 2 \times 10}$$

$$g_{m4} \Rightarrow 164 \mu\text{A}$$

$$\frac{(\frac{\omega}{L})_6}{(\frac{\omega}{L})_4} \Rightarrow \frac{1400 \mu\text{A}}{164 \mu\text{A}} \approx 8.57$$

$$(\frac{\omega}{L})_6 = (\frac{\omega}{L})_4 \times 8.57$$

$$\boxed{(\frac{\omega}{L})_6 \approx 120}$$

$$\frac{(\frac{\omega}{L})_6}{(\frac{\omega}{L})_4} = \frac{I_6}{I_4} \Rightarrow \frac{120}{14} \Rightarrow \frac{I_6}{10}$$

$$\boxed{\frac{1200}{14} = I_6 \approx 86 \mu\text{A} \approx I_7}$$

* For M7

$$\frac{(\frac{\omega}{L})_7}{(\frac{\omega}{L})_5} = \frac{I_7}{I_5}$$

$$\frac{(\frac{\omega}{L})_7}{12} = \frac{86 \mu\text{A}}{20 \mu\text{A}}$$

$$(\frac{\omega}{L})_7 \approx \underline{\underline{50}}$$

power dissipation $\Rightarrow P = VI$

$$\Rightarrow (V_{dd}) (86\mu + 20\mu)$$

$$\Rightarrow 1.8 [106\mu A]$$

$$\text{power} \Rightarrow 190 \mu W \leq 300 \mu W$$

obtained in Cadence virtuoso.

$$\text{Gain} \Rightarrow 61 \text{ dB}$$

$$\text{phase margin} \Rightarrow 44.75^\circ$$

$$\text{power dissipated} = 206 \mu W \leq 300 \mu W$$

